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A Study on Methods, Challenges, and Future Directions of Generative Adversarial Networks in Stock Market Prediction

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***Abstract*—Generative Adversarial Networks (GANs) have demonstrated a transformative capacity in financial modeling, particularly stock price prediction. This systematic literature review delves into the methodologies, challenges, and efficacy of GANs in this field, underscoring their ability to generate complex, high-dimensional data that traditional models struggle to process. By learning from the distribution of historical stock data, GANs can uncover intricate patterns missed by conventional approaches, offering a more nuanced understanding of market dynamics. However, their deployment is not without challenges, including training instability and the complex nature of financial data, which can lead to issues like mode collapse and non-convergence. The review aggregates insights from various studies to provide a comprehensive overview of current applications, highlight innovative methods, and outline future research directions to further harness GANs' potential in financial analytics. The ultimate aim is to illuminate how GANs could revolutionize traditional stock market analytics, transforming them into more predictive, robust tools for investors and analysts.**

Keywords— generative adversarial networks (GANs), stock price prediction, financial forecasting, deep learning in finance, predictive analytics

I. INTRODUCTION

Generative Adversarial Networks (GANs) have transformed the field of artificial intelligence by generating complicated, high-dimensional data [1]. GANs have shown great potential in financial modeling, notably stock price prediction [2]. This expertise is critical in an industry where accurate stock price forecasting has a substantial impact on strategic investment decisions and risk management [3]. GANs, which analyze the distribution of past stock data, can find detailed patterns that traditional models may miss, providing a more dynamic approach to interpreting market movements [4].

Historically, stock price forecasting has depended primarily on statistical and machine learning models like Autoregressive Integrated Moving Average (ARIMA), Support Vector Machines, and deep learning techniques like Long Short-Term Memory (LSTM) networks [5]. While these methodologies have yielded valuable insights, they frequently fail to capture the nonlinear intricacies of financial markets [6]. GANs established a new paradigm, allowing for improved predictive accuracy and the production of synthetic financial time series that reflect probable future situations, so providing a more robust foundation for evaluating financial strategies and models [7].

GANs were introduced by Ian Goodfellow and his colleagues in 2014 [8]. Figure 1 shows that adversarial learning trains the generator and discriminator neural networks simultaneously. From random noise, the generator generates synthetic data that replicates real data. It aims to provide data that the discriminator cannot differentiate from genuine data. The discriminator compares real and synthetic data from the generator. Its purpose is to appropriately recognize real or fabricated input data.

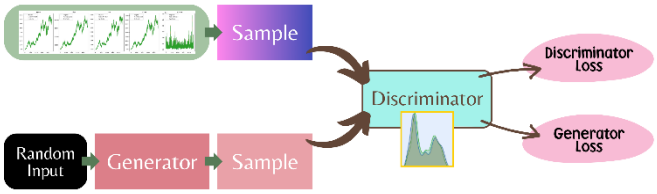


Fig. 1. GAN architecture

The training process involves a minimax game, where the generator aims to maximize the probability of the discriminator making a mistake, while the discriminator aims to minimize this probability. Mathematically, the objective function can be expressed as:

$$\min_G \max_D V(D,G) = E_{x\sim p_{data}(x)}[\log D(x)] + E_{z\sim p_z(z)}\left[\log\left(1 - D\left(G(z)\right)\right)\right] \quad (1)$$

Where $D(x)$ is the discriminator's estimate of the probability that real data x is real; $G(z)$ is the generator's output when given noise z; $p_{data}(x)$ is the real data

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distribution; and $p_z(z)$ is the noise distribution. This adversarial process continues until the discriminator can no longer distinguish between real and synthetic data, indicating that the generator is producing highly realistic data.

In this research, Generative Adversarial Networks (GANs) are utilized as the core technology to address the complexities of stock market prediction. The study showed how GANs can improve prediction accuracy by learning the data distribution [4]. Hybrid models using GANs and other architectures like LSTM networks have improved stock price temporal dynamics [6]. These developments make GANs perfect for nonlinear and volatile financial markets, improving investor predictions and strategic insights. Advanced models like Wasserstein GAN (WGAN) and its modifications improve model stability and provide high-quality synthetic data, addressing GAN concerns such mode collapse [9].

This paper presents significant contributions to GAN-based financial modeling and predictive analytics, offering a comprehensive synthesis of existing research on the use of GANs in stock market prediction. It addresses key challenges and solutions, providing readers with a deeper understanding of GAN applications in this domain. By exploring innovative methods such as probabilistic forecasting and hybrid models, the paper highlights promising approaches to enhance GAN performance in financial forecasting. Furthermore, it discusses practical applications in financial data generation, risk management, and strategic decision-making, offering valuable insights for improving prediction accuracy and strategic planning. The paper also outlines future research directions to overcome current limitations, paving the way for the development of more advanced and reliable predictive models. Ultimately, this work aims to demonstrate the transformative potential of GANs in revolutionizing stock market analytics, making them more predictive and robust tools for investors and analysts.

II. Methodology

A. Search Strategy and Selection Criteria

The methodology of this study was specifically designed to evaluate the effectiveness of GANs in predicting stock prices. An exhaustive search was conducted using the Scopus database, known for its extensive repository of peer-reviewed articles. The search utilized a precise keyword strategy to ensure the inclusion of all relevant literature on the subject. The keywords used were: ("Generative Adversarial Network" OR "GAN") AND ("stock" OR "financial time series") AND ("prediction" OR "forecasting" OR "market" OR "price" OR "synthetic" OR "augmented" OR "augmentation").

The selection process involved several stages to refine the pool of articles. Initially, 161 publications were identified, which were then narrowed down to 144 after applying a date filter restricting the scope to publications from 2019 to 2024 to ensure the relevance and recency of the data.

B. Screening and Eligibility

The selection process for this SLR was meticulously structured to assess the efficacy of GANs in financial forecasting. Initially, the articles were identified and subjected to a preliminary screening of titles and abstracts, focusing on their relevance to applying GANs in stock price prediction. This was followed by a detailed review of the full texts, emphasizing the studies' methodologies and alignment with specific research questions, which refined the selection to 53. The final inclusion of 34 publications was determined based on the depth of analysis, methodological rigor, and their direct relevance to GANs' application in financial forecasting. This rigorous process, depicted in Fig. 2, aimed to compile a comprehensive synthesis of GANs' capabilities and advancements, ensuring a robust foundation for this review. The criteria for exclusion included non-English articles, conference abstracts lacking full papers, and studies not specifically focusing on GANs for financial time series analysis.

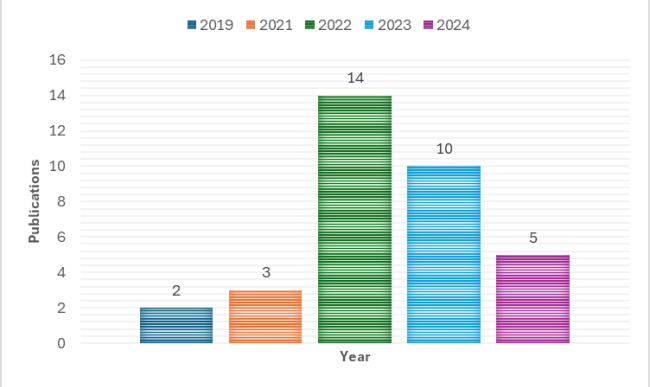


Fig. 2. Number of publications based on year

C. Data Extraction Process

For each eligible study, detailed information was extracted, including the authors, year of publication, the geographical context, the dataset used, the specific type of GAN implemented, and key findings related to the effectiveness and challenges of using GANs in stock market predictions. Special attention was given to variations in GAN architecture, such as Conditional GANs, Wasserstein GANs with Gradient Penalty, and their unique applications in stock price forecasting. This data extraction allowed for a comprehensive analysis of current methodologies and outcomes across different studies.

D. Quality Assessment

To ensure the reliability and validity of the findings, a quality assessment was conducted on each selected study. This assessment focused on the robustness of the research methodology, the clarity of the data presentation, and the reproducibility of the results. Studies that demonstrated high methodological quality, clear articulation of outcomes, and significant contributions to understanding GAN applications in stock price prediction were weighted more heavily in the synthesis of this review.

E. Synthesis of Findings

The final step involved synthesizing the data extracted from the selected studies to identify common themes, trends, and gaps in the research landscape. The synthesis highlighted the diverse applications and configurations of GANs and underscored the common challenges faced, such as model stability and overfitting. This structured analysis provided a grounded basis for discussing the current state of GAN technology in financial forecasting and outlining potential future research directions that could address existing limitations and explore new possibilities within the domain.

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III. RESULTS

A. Overview of Studies

The rapidly evolving field of GANs in financial forecasting is well-represented in the diversity of models developed over the years. In Table I, we present a curated collection of pioneering GAN models, organized by the year of publication, showcasing the trajectory of innovation from 2019 to 2024. These models range from early adaptations for financial time-series analysis to sophisticated multi-source strategies and advanced network integrations. The evolution of these models reflects a deepening understanding of market dynamics and a persistent pursuit of enhanced predictive accuracy, with each year building upon the last to push the boundaries of what's possible in financial forecasting. The table serves as a testament to the field's ongoing research and development efforts, indicating a vibrant and forward-moving domain that leverages the power of GANs to anticipate and strategize within the ever-changing financial markets.

TABLE I. CHRONOLOGICAL DISTRIBUTION OF THE REVIEWED STUDIES

Year	Models
2019	GANs for financial time series [10], Deep Hierarchical Strategy Model for Multi-Source Driven Quantitative Investment [9]
2021	GAN with Gated Recurrent Units (GRU) and Convolutional Neural Networks (CNN) [11], GANs in the Generation of Stock Market Forecast Trend Images [12]
2022	StockGAN [13], Self-regulated GAN [14], Wasserstein GAN with Gradient Penalty (WGAN-GP) [15], WGAN-GP with XGBoost [16], GAN with LSTM and CNN [17], Hierarchical-SigCWGAN [18], MMGAN-HPA [19], GAN to evaluate information in financial markets [20], QuantumPath involving Feynman–Dirac path integrals, deep Bayesian networks, and t-GAN [21], Modified GANs [22], Deep Convolutional GAN (DCGAN) [23], DeepPricing [24], Quantum GAN (QGAN) and Quantum Circuit Born Machine (QCBM) [25]
2023	GAF-EWGAN [26], Multi-agent Virtual Market Model (MVMM) [27], RSQGAN [28], PredACGAN [29], ORGAN-FVT [30], GAN for predicting extreme market movements [31], AUDA-Net [32], GAN with PLR (Piecewise Linear Representation) [33], Modified QuantGAN with regime clustering [34]
2024	VGC-GAN [35], DRAGAN and Feature Matching [36], Fin-GAN [37], GAN-TrellisNet [38], GAN-Enhanced Nonlinear Fusion Model [39]

B. Enhancement of Prediction Accuracy

GANs significantly enhance the accuracy of stock price predictions through a multifaceted approach characterized by advanced modeling techniques, superior prediction performance, and targeted integrations with other technologies, as detailed in Table II. By utilizing complex architectures such as multi-graph convolutional networks and quantum GANs, GANs improve the robustness and stability of predictions, crucial for dealing with financial market volatilities. These networks excel in multi-step forecasting by integrating advanced neural configurations, which capture temporal dynamics and extreme market fluctuations more effectively than traditional models. This comprehensive enhancement in prediction capabilities makes GANs a powerful tool in financial forecasting, offering significant improvements over conventional predictive models.

TABLE II. GANS IN ENHANCING STOCK PRICE PREDICTION ACCURACY

Concept	Description	Reference
Model Techniques and Enhancements	GANs utilize advanced modeling techniques, such as multi-graph convolutional networks, quantum GANs, and hybrid architectures, to enhance prediction robustness and stability.	[10], [13], [14], [15], [16], [19], [23], [25], [35], [36], [37], [38], [39]
Prediction Performance	GANs improve prediction accuracy, especially in multi-step forecasting and extreme market conditions, demonstrating superior performance metrics over traditional models.	[11], [17], [22], [26], [29], [30], [31], [32]
Integration and Application Specific	GANs are integrated with technologies like XGBoost for feature optimization and are specifically applied to simulate market regimes and incorporate sentiment analysis effectively.	[10], [12], [14], [16], [20], [21], [24], [28], [34], [38], [39], [41]

C. Integration of Sentiment Analysis

Integrating sentiment analysis from social media and macroeconomic indicators into GAN models significantly improves the accuracy of financial market predictions. The inclusion of sentiment analysis enables the models to capture public perceptions and emotional responses often reflected in market movements, thereby enhancing prediction accuracy [11]. By integrating sentiment data directly, GANs can more effectively forecast market trends and adjust predictions based on real-time changes in public sentiment [41]. Moreover, incorporating social media sentiment analysis allows GANs to access a broader array of data inputs, providing a more holistic view of the factors influencing the markets [14]. Additionally, combining sentiment analysis with diverse data sources leads to a more comprehensive approach to market predictions, leveraging qualitative insights from sentiment data and quantitative signals from macroeconomic indicators [12]. This multifaceted integration enriches the model's dataset, leading to more accurate and robust financial market predictions.

D. Influence of Performance Metrics

Performance metrics like Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are essential in assessing the effectiveness of GANs in financial forecasting, as detailed in Table III. These metrics facilitate model evaluation and improvement by quantifying accuracy enhancements in GAN models, allowing for targeted optimizations and iterative improvements. They also determine model robustness, particularly under adverse market conditions, ensuring that GANs provide reliable and consistent outputs during market crashes and other stressful scenarios. Additionally, RMSE and MAE assess the data quality of synthetic data generated by GANs, verifying that it closely mimics real-world data and is suitable for robust financial analysis. These metrics are crucial for refining GANs, ensuring their reliability, and enhancing their practical application in predicting financial market trends and behaviors.

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TABLE III. ROLE OF PERFORMANCE METRICS IN EVALUATING GANS' EFFECTIVENESS

Concept	Description	Reference
Model Evaluation and Improvement	Performance metrics such as MSE, MAE, RMSE, and MAPE are essential for measuring the accuracy and efficiency improvements in GANs. They enable precise tracking and quantification of enhancements in model performance.	[11], [16], [17], [35], [38]
Model Robustness	Metrics like R^2 and MAE evaluate GAN robustness, particularly under stressful conditions such as market crashes, ensuring reliability and consistency in outputs even during extreme market conditions. Feature matching also plays a role in the robust training of GANs.	[13], [25], [26], [36]
Data Quality Assessment	Performance metrics are used to assess the fidelity and accuracy of synthetic data generated by GANs, comparing it against real data distributions to ensure high-quality outputs suitable for practical applications in financial forecasting.	[15], [23], [34]

E. Enhancing Decision-Making

GANs significantly enhance financial decision-making and strategic planning by generating and analyzing synthetic financial data, impacting key areas as outlined in Table IV. GANs improve risk management by allowing firms to simulate and prepare for various financial scenarios, effectively testing and refining risk mitigation strategies. They also support strategic decision-making by providing precise synthetic data that aids in forming comprehensive strategic plans. Furthermore, GANs boost predictive analysis and forecasting by enhancing the accuracy of financial predictions and developing probabilistic models. They deepen market understanding through realistic market simulations, enabling the testing of financial strategies in a controlled environment. Additionally, GANs are utilized in specialized applications like developing trading strategies for newly listed stocks where historical data is scarce. Collectively, these capabilities make GANs a pivotal technology in financial markets, facilitating more informed, accurate, and strategic operations.

TABLE IV. THE IMPACT OF GANS ON FINANCIAL DECISION-MAKING AND STRATEGIC PLANNING THROUGH SYNTHETIC DATA

Concept	Description	Reference
Risk Management and Control	GANs facilitate risk management by simulating various risk scenarios, allowing for effective testing and refinement of risk mitigation strategies.	[9], [26], [28], [29]
Strategic Decision-Making and Planning	GANs support strategic decision-making by generating accurate synthetic data, crucial for formulating comprehensive and informed strategic plans.	[11], [13], [16], [17], [18], [21], [34]
Predictive Analysis and Forecasting	GANs improve financial forecasts and develop probabilistic models that enhance the accuracy and reliability of predicting future market behaviors.	[12], [16], [19], [23], [27], [29], [37]

F. Challenges in Deployment

GANs for financial time series forecasting present challenges addressed through various strategic approaches, as detailed in Table V. Data quality and reliability involve rigorous validation and enhancement techniques to maintain accurate and comprehensive data inputs. The complexity of GAN models necessitates simplified architectures and user-friendly training procedures to reduce operational difficulties. Computational demands and scalability issues require efficient algorithms and hardware optimizations to manage large datasets effectively without excessive resource consumption. Additionally, adapting GANs to financial markets' dynamic and unpredictable nature involves continuous model updates and the swift integration of new data. Finally, achieving stability and precision in volatile market conditions is crucial, tackled by employing techniques to prevent training instability and mode collapse and conducting extensive testing across various scenarios to ensure consistent and reliable forecasts. These comprehensive strategies collectively enhance the effectiveness and reliability of GANs in financial forecasting applications.

TABLE V. CHALLENGES AND SOLUTIONS IN DEPLOYING GANS

Concept	Description	References
Data Quality and Reliability	Challenges include ensuring data completeness, accuracy, and integrity. Approaches involve rigorous data validation and enhancement to improve data quality used in GAN models.	[11], [12], [15], [21], [25], [27]
Model Complexity and Training Challenges	GANs require complex setups and precise tuning. Reducing operational complexity requires simplifying model architectures and developing user-friendly training procedures.	[9], [15], [20], [23], [35], [37]
Computational Demands and Scalability	The high computational demands of GANs necessitate optimizing computational algorithms and using efficient hardware. Scalability improvements are essential for handling large datasets without significant resource increases.	[9], [18], [25], [30], [35]

IV. DISCUSSION

A. Innovations and Challenges

GAN stock price prediction is innovative and difficult. Financial forecasting accuracy and training stability improve with multi-factor GANs [38]. In volatile markets, multi-graph convolutional networks can improve GAN model interpretability and performance [35]. Despite these gains, assessing prediction uncertainty in portfolio management is difficult, revealing a GAN application enhancement opportunity [29].

Modern GANs can better imitate complex financial time series, but model quality is still a challenge [34]. GANs' high computing requirements and accuracy issues in changing stock price circumstances make deployment difficult [13]. Another improvement area is GANs' failure to acquire stock market detail without feature engineering.

Real-time social media data integration into GAN models reveals data quality and relevance concerns, which affect model accuracy [14]. The issue of GANs' generality across financial instruments highlights the difficulty of adjusting these models to varied market settings [9]. Finally, classifying financial time series data, notably training data quality and model sensitivity, underlines the ongoing challenges to using GANs in financial forecasting [37].

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B. Strategic Implications

GANs may generate synthetic data that closely matches stock market trends, which has strategic significance in financial markets. This transformative skill enhances financial simulations, stress testing, and predictions. GAN integration with multiple modeling techniques improves financial analytics prediction [30]. Multi-factor GAN models and applications improve crisis forecasting and financial market resilience [26].

GANs test algorithms and build strong trading strategies by producing synthetic market scenarios. This application reduces risk and boosts earnings by generalizing Reinforcement Learning-based trading strategies [27]. GANs reliably predict financial market volatility, boosting AUC and F1 scores for financial decision-making. These fresh applications show GANs' potential to alter financial instrument market analysis and strategy development [13], [17].

GANs improve forecast accuracy by strategically transforming raw market data into predictive insights. GANs manage complex datasets and simulate market settings to enable financial systems adapt to rapidly changing market conditions [20]. To understand and prepare for market developments, they can evaluate complex dynamics and generate extreme market data.

C. Future Directions

GAN stock price prediction research uses new deep learning models, regularization approaches, and cross-market data to increase flexibility and accuracy. GANs can adapt to different data inputs and enhance financial forecasts by turning raw pricing data into predictive insights [20].

Over earlier methodologies, multi-factor GAN models improve stock price knowledge and progress [38]. Using GANs and other robust models during market crises helps maintain profitability [26]. Merging GANs with LSTM and CNN for directional stock price prediction improved accuracy and efficiency, proving hybrid methods in complex predictive situations [17].

Final research on GANs for accurate market projections. Financial volatility trend classification GANs improve prediction. ARIMA and GAN frameworks improve stock price predictions by capturing and analyzing temporal trends. These methods show the ongoing effort to increase GAN accuracy and actionability, enabling more advanced trading strategies and financial analysis [27].

V. Conclusion

Generative Adversarial Networks (GANs) represent a significant advancement in predictive analytics for the stock market by offering sophisticated tools that surpass traditional models in handling the complexity and non-linearity of financial data. By learning the underlying distributions of stock prices, GANs can generate new data instances that mimic authentic stock market behavior, thereby offering valuable insights into potential future trends. This capability enhances predictive accuracy and provides a richer basis for testing financial strategies and models, making GANs a powerful asset in financial forecasting.

Despite these advantages, the full potential of GANs in stock market prediction remains unrealized due to ongoing challenges in training stability, such as mode collapse and non-convergence, which can hinder practical implementation. The inherent noise and non-stationary nature of financial data add further complexity to GAN training. Thus, further research is necessary to explore the theoretical foundations of GANs to address these training challenges and improve model robustness. Practical implementations also require refinement to ensure reliable performance across varying market conditions. Additionally, regulatory implications of using advanced predictive tools in financial markets must be carefully considered to ensure transparency and fairness. Future research should focus on integrating newer deep learning models, enhanced regularization techniques, and cross-market analysis to boost GAN adaptability and accuracy, ultimately transforming stock market analytics into more predictive and robust tools for investors and analysts.

References

[1] I. Goodfellow et al., “Generative Adversarial Networks,” Communications of the Acm, 2020, doi: 10.1145/3422622.

[2] D. Saxena and J. Cao, “Generative Adversarial Networks (GANs),” Acm Computing Surveys, 2021, doi: 10.1145/3446374.

[3] H. Alqahtani, M. Kavakli, and G. Kumar, “Applications of Generative Adversarial Networks (GANs): An Updated Review,” Archives of Computational Methods in Engineering, 2019, doi: 10.1007/s11831-019-09388-y.

[4] J. Gui, Z. Sun, Y. Wen, D. Tao, and J. Ye, “A Review on Generative Adversarial Networks: Algorithms, Theory, and Applications,” IEEE Transactions on Knowledge and Data Engineering, 2023, doi: 10.1109/tkde.2021.3130191.

[5] D. Saxena and J. Cao, “Generative Adversarial Networks (GANs),” Acm Computing Surveys, 2021, doi: 10.1145/3446374.

[6] H. Alqahtani, M. Kavakli, and G. Kumar, “Applications of Generative Adversarial Networks (GANs): An Updated Review,” Archives of Computational Methods in Engineering, 2019, doi: 10.1007/s11831-019-09388-y.

[7] J. Gui, Z. Sun, Y. Wen, D. Tao, and J. Ye, “A Review on Generative Adversarial Networks: Algorithms, Theory, and Applications,” IEEE Transactions on Knowledge and Data Engineering, 2023, doi: 10.1109/tkde.2021.3130191.

[8] M. M. Sharaf, E. E. Hemdan, A. El‐Sayed, and N. A. El‐Bahnasawy, “StockPred: A Framework for Stock Price Prediction,” Multimedia Tools and Applications, 2021, doi: 10.1007/s11042-021-10579-8.

[9] A. H. Bukhari, M. A. Z. Raja, M. Sulaiman, S. Islam, M. Shoaib, and P. Kumam, “Fractional Neuro-Sequential ARFIMA-LSTM for Financial Market Forecasting,” IEEE Access, vol. 8, pp. 71326–71338, 2020, doi: 10.1109/ACCESS.2020.2985763.

[10] M. Jiang, L. Jia, Z. Chen, and W. Chen, “The Two-Stage Machine Learning Ensemble Models for Stock Price Prediction by Combining Mode Decomposition, Extreme Learning Machine and Improved Harmony Search Algorithm,” Annals of Operations Research, 2020, doi: 10.1007/s10479-020-03690-w.

[11] G. Biau, B. Cadre, M. Sangnier, and U. Tanielian, “Some Theoretical Properties of GANS,” The Annals of Statistics, 2020, doi: 10.1214/19-aos1858.

[12] C. Tang, W. Zhu, and X. Yu, “Deep hierarchical strategy model for multi-source driven quantitative investment,” IEEE Access, vol. 7, pp. 79331–79336, 2019, doi: 10.1109/ACCESS.2019.2923267.

[13] S. Takahashi, Y. Chen, and K. Tanaka-Ishii, “Modeling Financial Time-Series With Generative Adversarial Networks,” Physica a Statistical Mechanics and Its Applications, 2019, doi: 10.1016/j.physa.2019.121261.

[14] H. C. Lin, C. Chen, G. F. Huang, and A. Jafari, “Stock Price Prediction using Generative Adversarial Networks,” Journal of Computer Science, vol. 17, no. 3, 2021, doi: 10.3844/JCSSP.2021.188.196.

[15] D. Xiao, “Research on the Application of Generative Adversarial Networks in the Generation of Stock Market Forecast Trend Images,” Scientific Programming, vol. 2021, pp. 1–12, Dec. 2021, doi: 10.1155/2021/7321671.

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[16] M. Diqi, M. E. Hiswati, and A. S. Nur, “StockGAN: robust stock price prediction using GAN algorithm,” Int. J. Inf. Technol., vol. 14, no. 5, pp. 2309–2315, 2022, doi: 10.1007/s41870-022-00929-6.

[17] H. Xu, D. Cao, and S. Li, “A self-regulated generative adversarial network for stock price movement prediction based on the historical price and tweets,” Knowl Based Syst, vol. 247, 2022, doi: 10.1016/j.knosys.2022.108712.

[18] F. Han, X. Ma, and J. Zhang, “Simulating Multi-Asset Classes Prices Using Wasserstein Generative Adversarial Network: A Study of Stocks, Futures and Cryptocurrency,” J. Risk. Financ. Manag., vol. 15, no. 1, 2022, doi: 10.3390/jrfm15010026.

[19] J. Xu et al., “Financial Time Series Prediction Based on XGBoost and Generative Adversarial Networks,” International Journal of Circuits, Systems and Signal Processing, vol. 16, pp. 637–645, Jan. 2022, doi: 10.46300/9106.2022.16.79.

[20] A. Kumar et al., “Generative adversarial network (GAN) and enhanced root mean square error (ERMSE): deep learning for stock price movement prediction,” Multimed Tools Appl, vol. 81, no. 3, pp. 3995–4013, Jan. 2022, doi: 10.1007/s11042-021-11670-w.

[21] F. M. Pardo, P. Schwendner, and M. Wunsch, “Tackling the Exponential Scaling of Signature-Based Generative Adversarial Networks for High-Dimensional Financial Time-Series Generation,” J. Financial. Data. Sci., vol. 4, no. 4, pp. 110–132, 2022, doi: 10.3905/jfds.2022.1.109.

[22] S. R. Polamuri, K. Srinivas, and A. K. Mohan, “Multi-Model Generative Adversarial Network Hybrid Prediction Algorithm (MMGAN-HPA) for Stock Market Prices Prediction,” Journal of King Saud University - Computer and Information Sciences, 2022, doi: 10.1016/j.jksuci.2021.07.001.

[23] D. Santoro and L. Grilli, “Generative Adversarial Network to evaluate quantity of information in financial markets,” Neural Comput. Appl., 2022, doi: 10.1007/s00521-022-07401-3.

[24] F. Soleymani and E. Paquet, “Long-term financial predictions based on Feynman–Dirac path integrals, deep Bayesian networks and temporal generative adversarial networks,” Machine Learning with Applications, vol. 7, p. 100255, Mar. 2022, doi: 10.1016/j.mlwa.2022.100255.

[25] G. Sornavalli, G. Angelin, and N. H. Khanna, “Intelligent Forecast of Stock Markets to Handle COVID-19 Economic Crisis by Modified Generative Adversarial Networks,” Comput J, vol. 65, no. 12, pp. 3250–3264, 2022, doi: 10.1093/comjnl/bxac056.

[26] A. Staffini, “Stock Price Forecasting by a Deep Convolutional Generative Adversarial Network,” Front. Artif. Intell., vol. 5, p. 837596, Feb. 2022, doi: 10.3389/frai.2022.837596.

[27] X. Tan, Z. Zhang, X. Zhao, and S. Wang, “DeepPricing: pricing convertible bonds based on financial time-series generative adversarial networks,” Financial Innov., vol. 8, no. 1, 2022, doi: 10.1186/s40854-022-00369-y.

[28] E. Y. Zhu et al., “Generative quantum learning of joint probability distribution functions,” Phys. Rev. Res., vol. 4, no. 4, 2022, doi: 10.1103/PhysRevResearch.4.043092.

[29] A. Ghasemieh and R. Kashef, “An enhanced Wasserstein generative adversarial network with Gramian Angular Fields for efficient stock market prediction during market crash periods,” Appl Intell, vol. 53, no. 23, pp. 28479–28500, 2023, doi: 10.1007/s10489-023-05016-2.

[30] F.-F. He, C.-T. Chen, and S.-H. Huang, “A multi-agent virtual market model for generalization in reinforcement learning based trading strategies,” Appl. Soft Comput., vol. 134, 2023, doi: 10.1016/j.asoc.2023.109985.

[31] A. Huang, M. Khushi, and B. Suleiman, “Regime-Specific Quant Generative Adversarial Network: A Conditional Generative Adversarial Network for Regime-Specific Deepfakes of Financial Time Series,” Appl. Sci., vol. 13, no. 19, 2023, doi: 10.3390/app131910639.

[32] J. Kim and M. Lee, “Portfolio optimization using predictive auxiliary classifier generative adversarial networks,” Eng Appl Artif Intell, vol. 125, 2023, doi: 10.1016/j.engappai.2023.106739.

[33] L. Liu et al., “An Efficient GAN-Based Multi-classification Approach for Financial Time Series Volatility Trend Prediction,” Int. J. Comput. Intell. Syst., vol. 16, no. 1, 2023, doi: 10.1007/s44196-023-00212-x.

[34] B. Labiad, A. Berrado, and L. Benabbou, “Predicting extreme events in the stock market using generative adversarial networks,” Int. J. Adv. Intell. Informatics, vol. 9, no. 2, p. 218, Jul. 2023, doi: 10.26555/ijain.v9i2.898.

[35] Q. Wei and Q. Dai, “Adversarial unsupervised domain adaptation based on generative adversarial network for stock trend forecasting,” Intell. Data Anal., vol. 27, no. 5, pp. 1477–1502, 2023, doi: 10.3233/IDA-220414.

[36] J.-L. Wu, X.-R. Tang, and C.-H. Hsu, “A prediction model of stock market trading actions using generative adversarial network and piecewise linear representation approaches,” Soft Comput., vol. 27, no. 12, pp. 8209–8222, 2023, doi: 10.1007/s00500-022-07716-2.

[37] K. Zakharov, E. Stavinova, and A. Boukhanovsky, “Synthetic Financial Time Series Generation with Regime Clustering,” J. Adv. Inf. Technol., vol. 14, no. 6, pp. 1372–1381, 2023, doi: 10.12720/jait.14.6.1372-1381.

[38] D. Ma, D. Yuan, M. Huang, and L. Dong, “VGC-GAN: A multi-graph convolution adversarial network for stock price prediction,” Expert Sys Appl, vol. 236, 2024, doi: 10.1016/j.eswa.2023.121204.

[39] F. S. Nejad and M. M. Ebadzadeh, “Stock market forecasting using DRAGAN and feature matching,” Expert Sys Appl, vol. 244, 2024, doi: 10.1016/j.eswa.2023.122952.

[40] M. Vuletić, F. Prenzel, and M. Cucuringu, “Fin-GAN: forecasting and classifying financial time series via generative adversarial networks,” Quant. Financ., vol. 24, no. 2, pp. 175–199, 2024, doi: 10.1080/14697688.2023.2299466.

[41] W. Liu, Y. Ge, and Y. Gu, “Multi-factor stock price prediction based on GAN-TrellisNet,” Knowl. Inf. Systems. Syst., 2024, doi: 10.1007/s10115-024-02085-8.

[42] Y. Xu, Y. Zhang, P. Liu, Q. Zhang, and Y. Zuo, “GAN-Enhanced Nonlinear Fusion Model for Stock Price Prediction,” Int. J. Comput. Intell. Syst., vol. 17, no. 1, 2024, doi: 10.1007/s44196-023-00394-4.

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